

Sequence Quality Control (QC) and Trimming

- Homework #1 is due today at 11:59pm via Blackboard upload
- Homework #2 is on Blackboard Due Feb. 11th
- Open 03_Working_with_Sequences.qmd (same as on Tuesday)
- Run code chunks through line 245 (where we ended Tuesday)
- For personal computers, install FastQC and MultiQC (if you haven't done so already)

Levels of QC

- 1) RNA quality (undegraded and of sufficient quantity)
- 2) Sequencing quality (the focus of today's lecture)
- 3) Post-processing QC (future lectures)
 - mapping quality
 - replicate agreement

First QC Step: RNA Quality

Gallego Romero *et al. BMC Biology* 2014, **12**:42
<http://www.biomedcentral.com/1741-7007/12/42>



RESEARCH ARTICLE

Open Access

RNA-seq: impact of RNA degradation on transcript quantification

Irene Gallego Romero¹, Athma A Pai^{1,2}, Jenny Tung^{1,3} and Yoav Gilad^{1*}

Conclusions: While standard normalizations failed to account for the effects of degradation, we found that by explicitly controlling for the effects of RIN using a linear model framework we can correct for the majority of these effects. We conclude that in instances in which RIN and the effect of interest are not associated, this approach can help recover biologically meaningful signals in data from degraded RNA samples.

RNA Integrity Number (RIN)

BMC Molecular Biology



Methodology article

Open Access

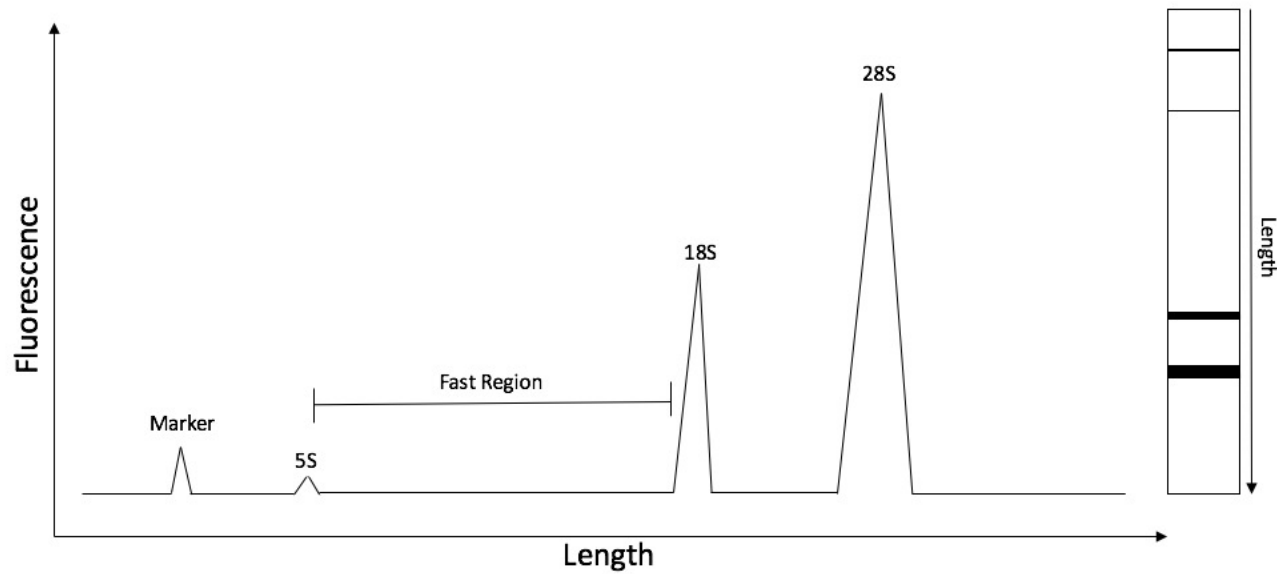
The RIN: an RNA integrity number for assigning integrity values to RNA measurements

Andreas Schroeder^{1,3}, Odilo Mueller^{*2}, Susanne Stocker^{1,4},
Ruediger Salowsky¹, Michael Leiber¹, Marcus Gassmann¹, Samar Lightfoot²,
Wolfram Menzel⁵, Martin Granzow⁶ and Thomas Ragg^{*6}

Basic principle:

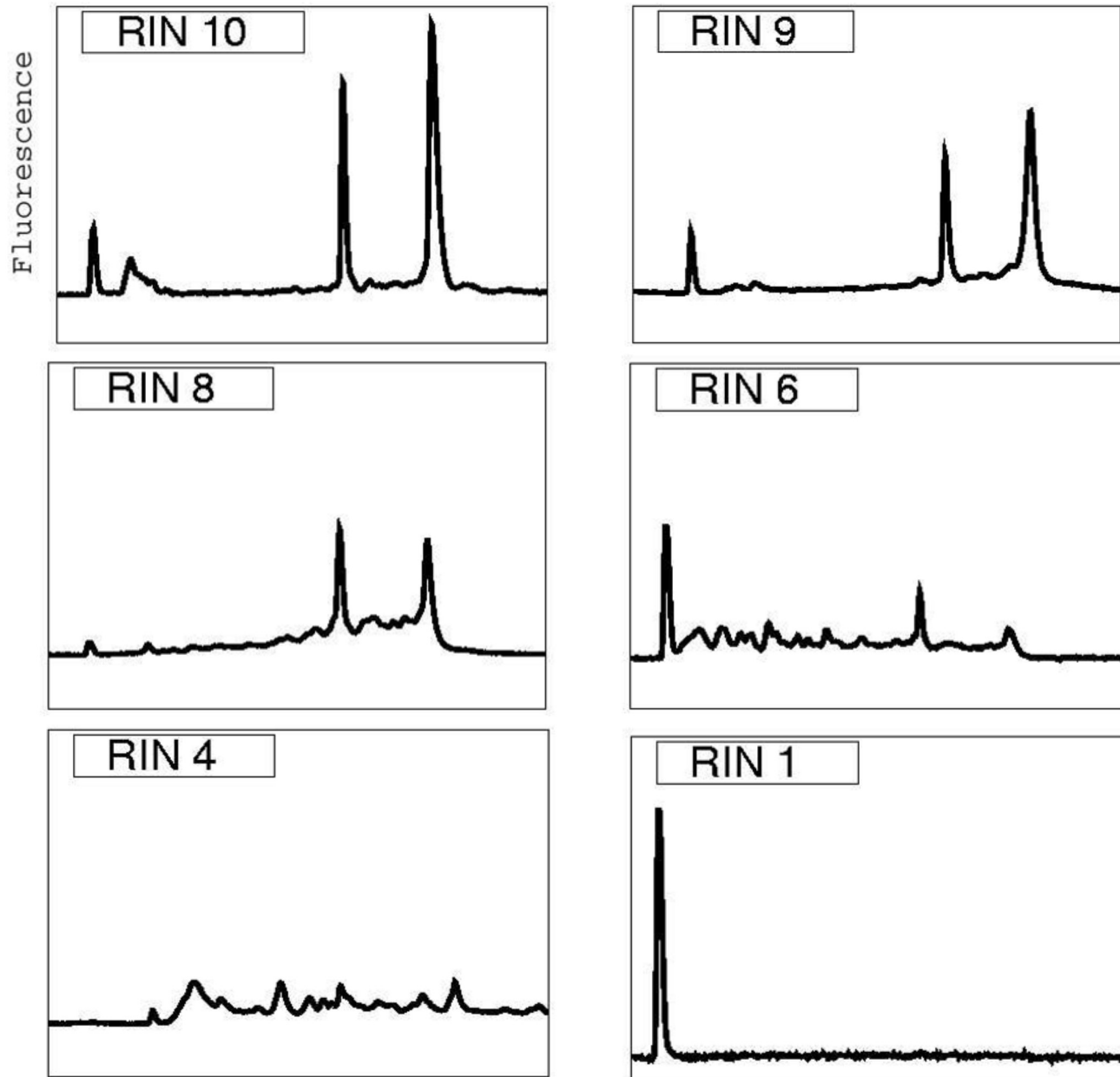
- eukaryotic rRNA (the dominant species in total RNA) consists of 28S, 18S, and 5S subunits in specific proportions
- 28S, 18S, and 5S RNA do not degrade at the same rate, so deviations in the ratio of each can be used to quantify RNA quality
- RIN is calculated by an automated algorithm on a 1-10 scale (10 being fully intact and 1 being fully degraded)

RNA Integrity Number (RIN)



Idealized electrophoretic trace of intact RNA (RIN = 10)

RNA Integrity Number (RIN)



FASTQC



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FastQC

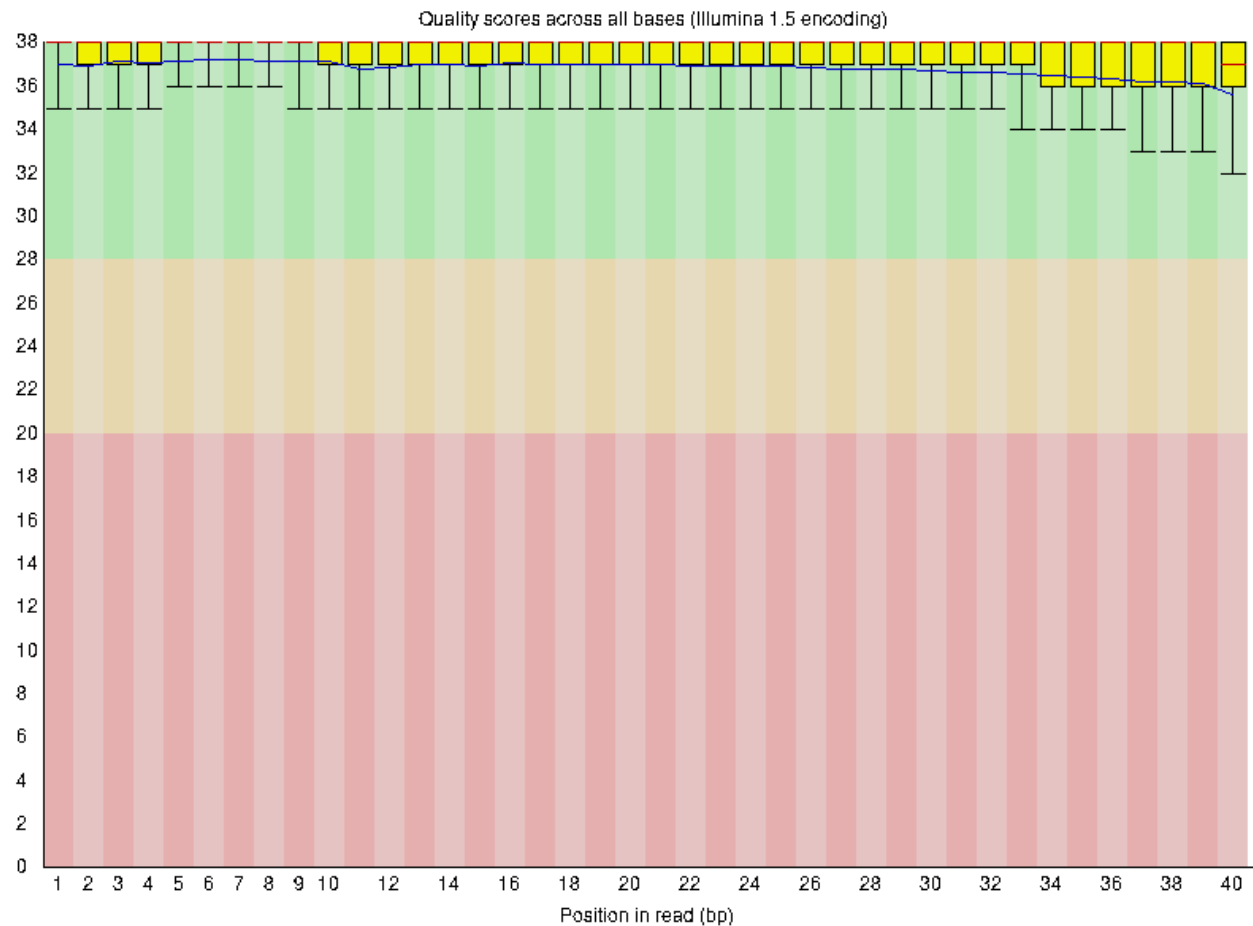
Function	A quality control tool for high throughput sequence data.
Language	Java
Requirements	A suitable Java Runtime Environment The Picard BAM/SAM Libraries (included in download)
Code Maturity	Stable. Mature code, but feedback is appreciated.
Code Released	Yes, under GPL v3 or later .
Initial Contact	Simon Andrews
Download Now	

Common Problems Identified in FASTQC

- <https://sequencing.qcfail.com/> has a number of great posts on how sequencing can fail

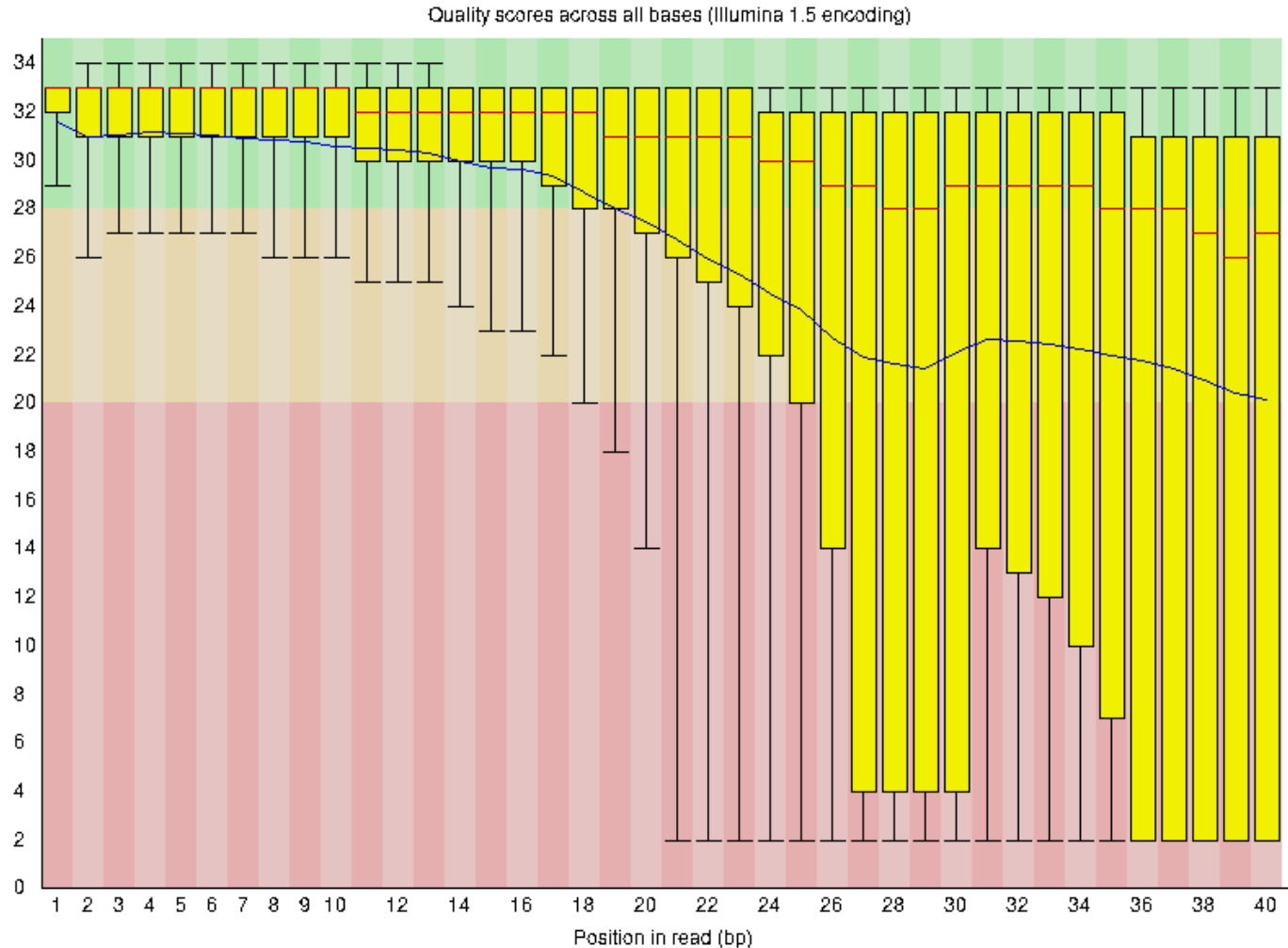
Common Problems Identified in FASTQC

Good: Phred scores mostly >30



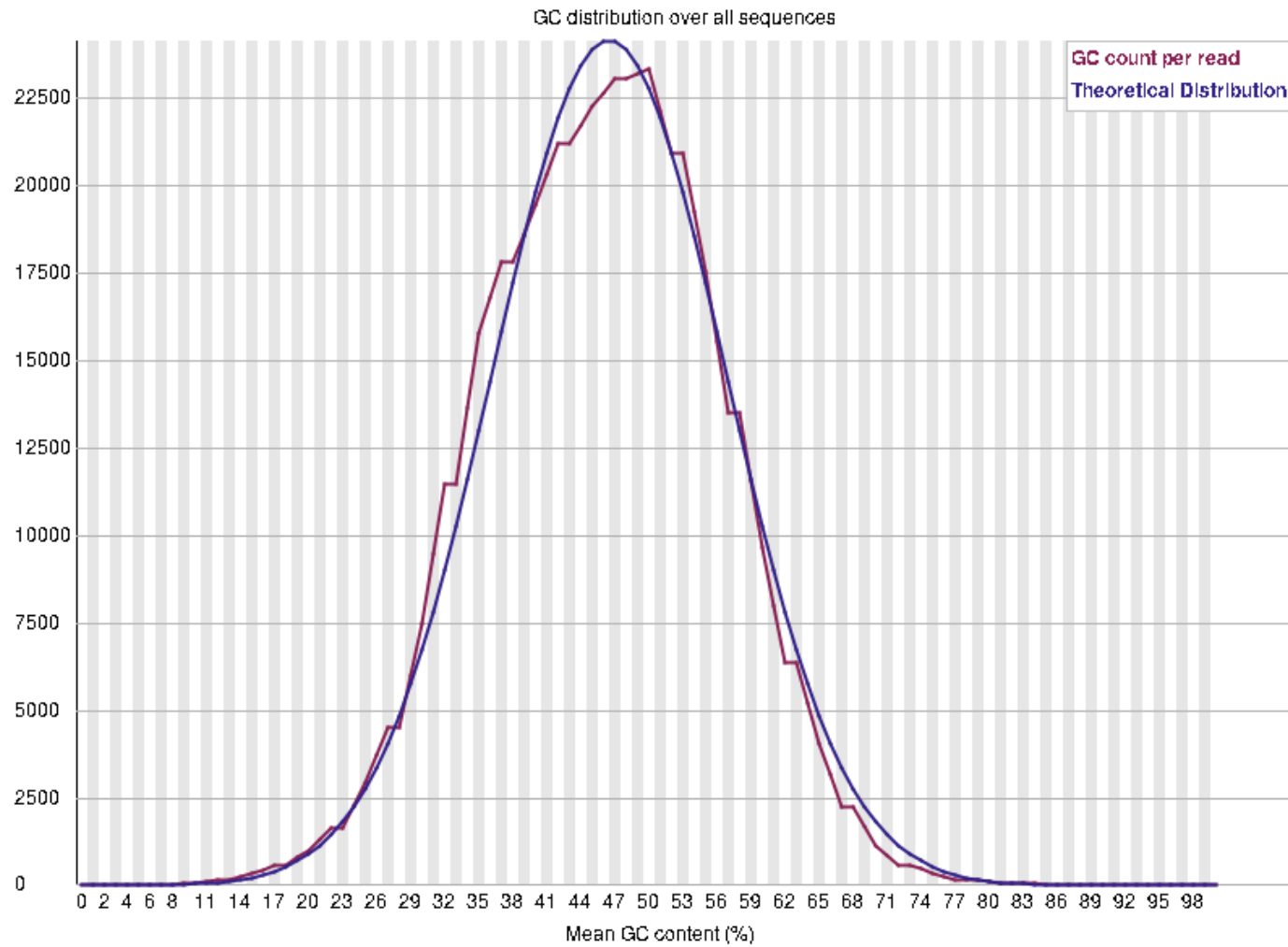
Common Problems Identified in FASTQC

Bad: sequence quality drops off (could be run failure or sample issue)



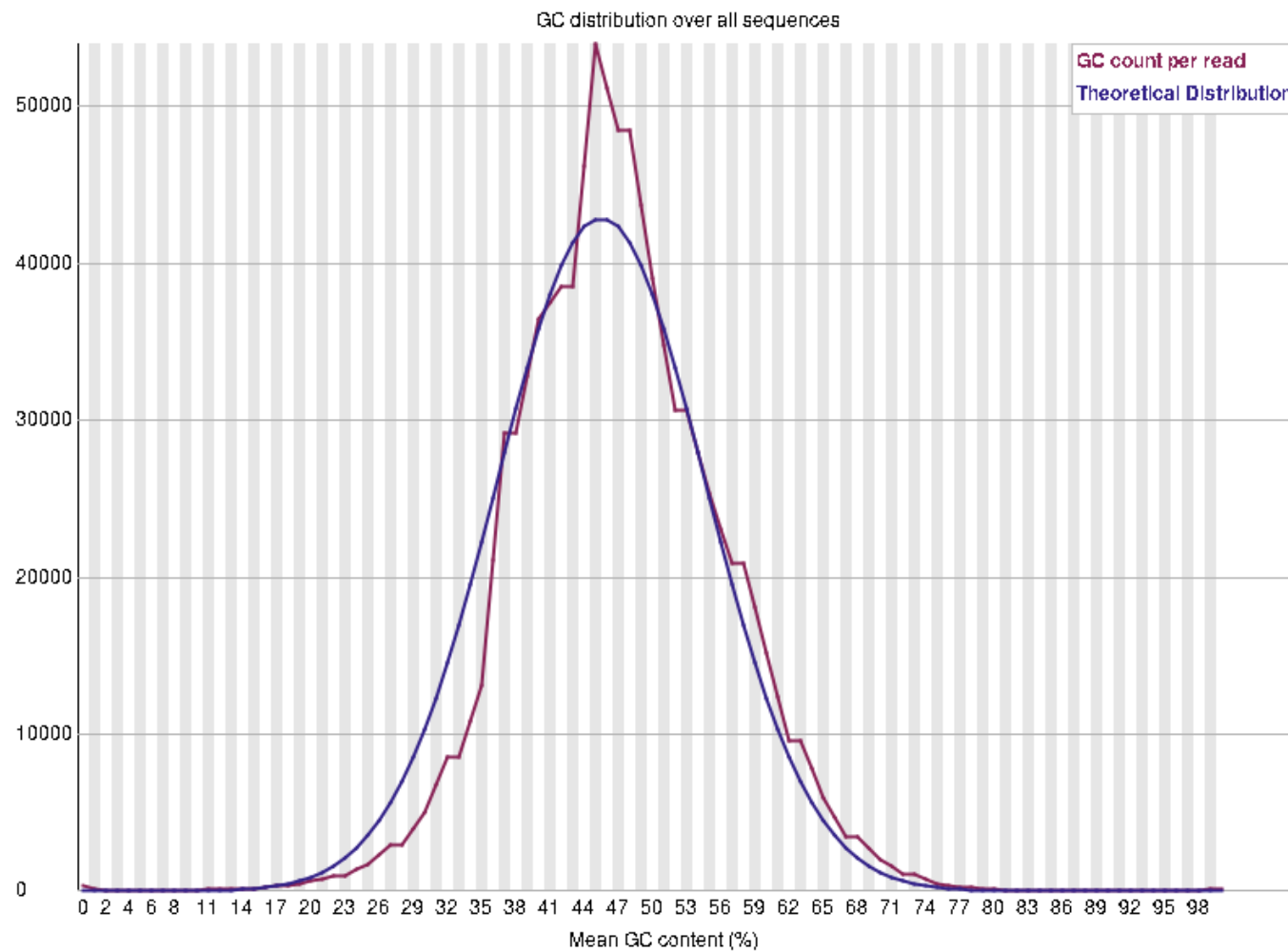
Common Problems Identified in FASTQC

Good



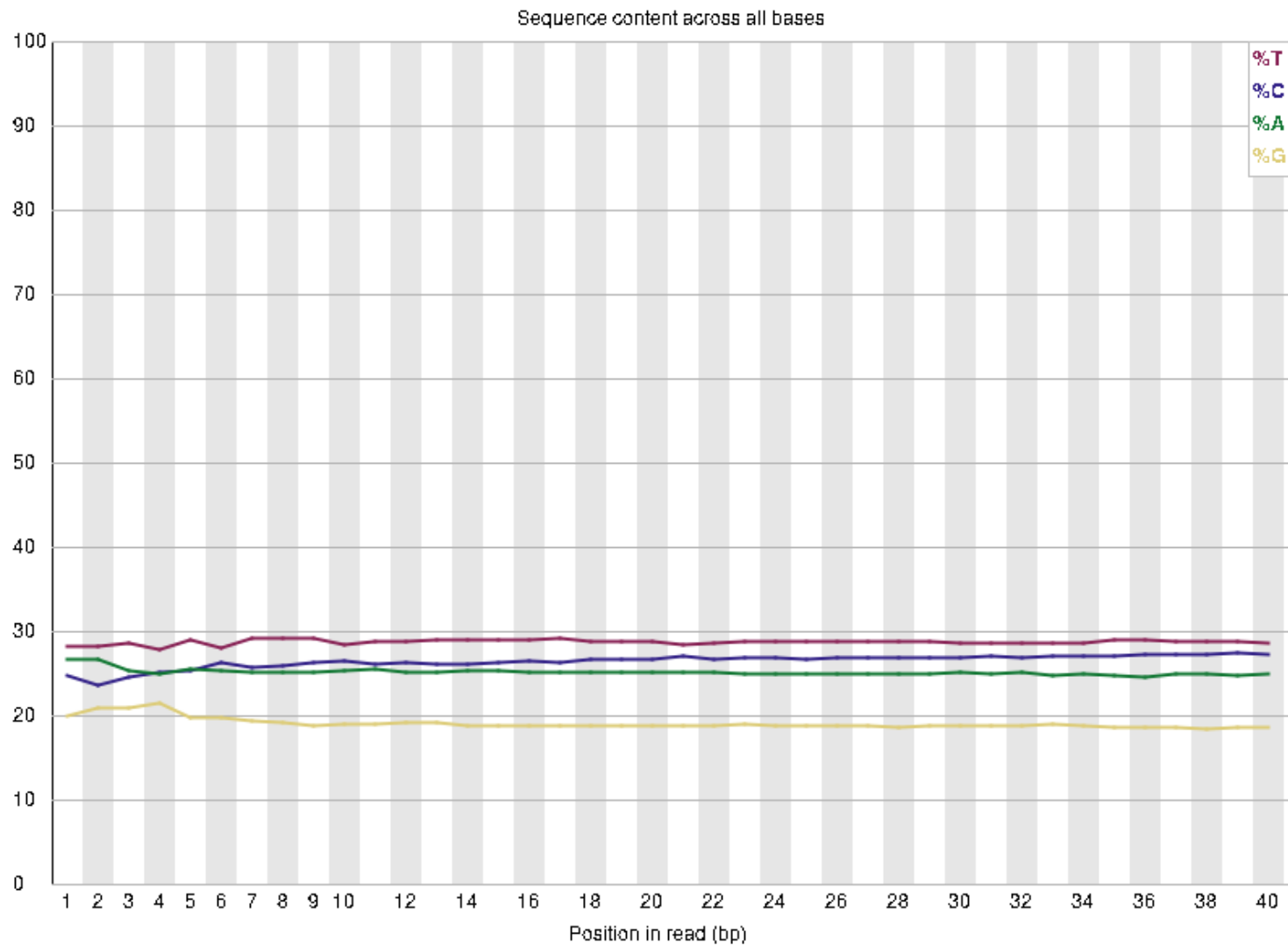
Common Problems Identified in FASTQC

Bad: non uniform could indicate contamination from another organism



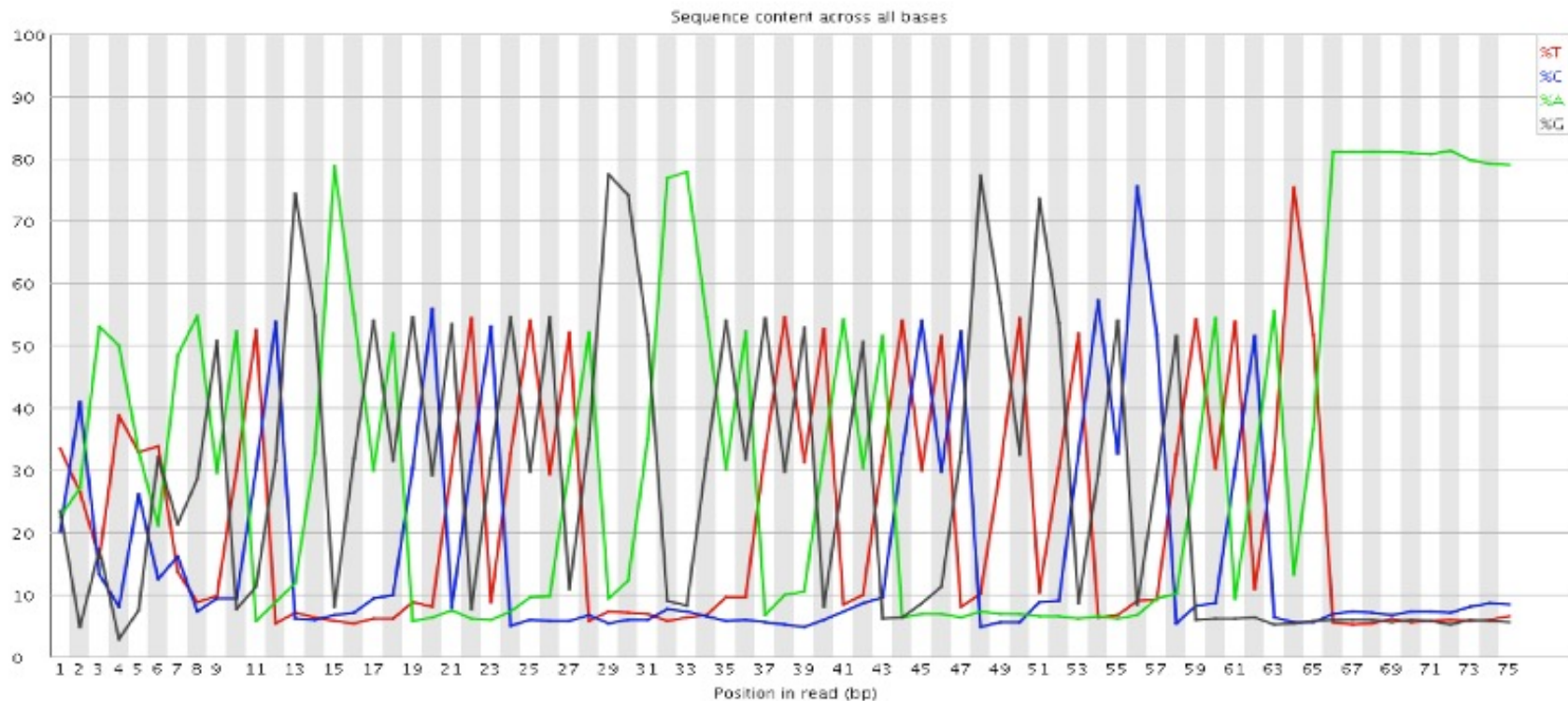
Common Problems Identified in FASTQC

Good



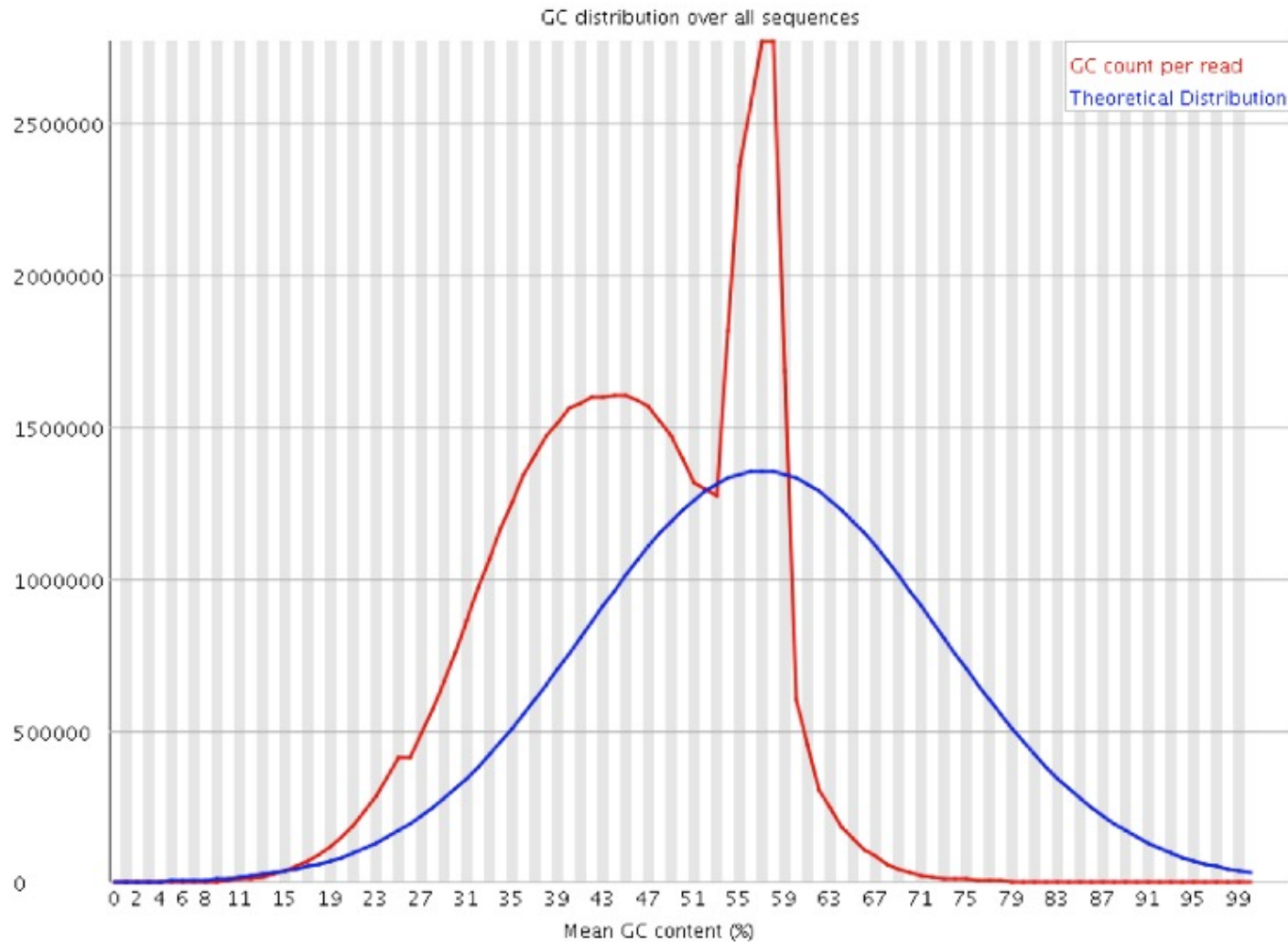
Common Problems Identified in FASTQC

Bad: major adapter contamination
(would show up on GC distribution),
non-diverse library



Common Problems Identified in FASTQC

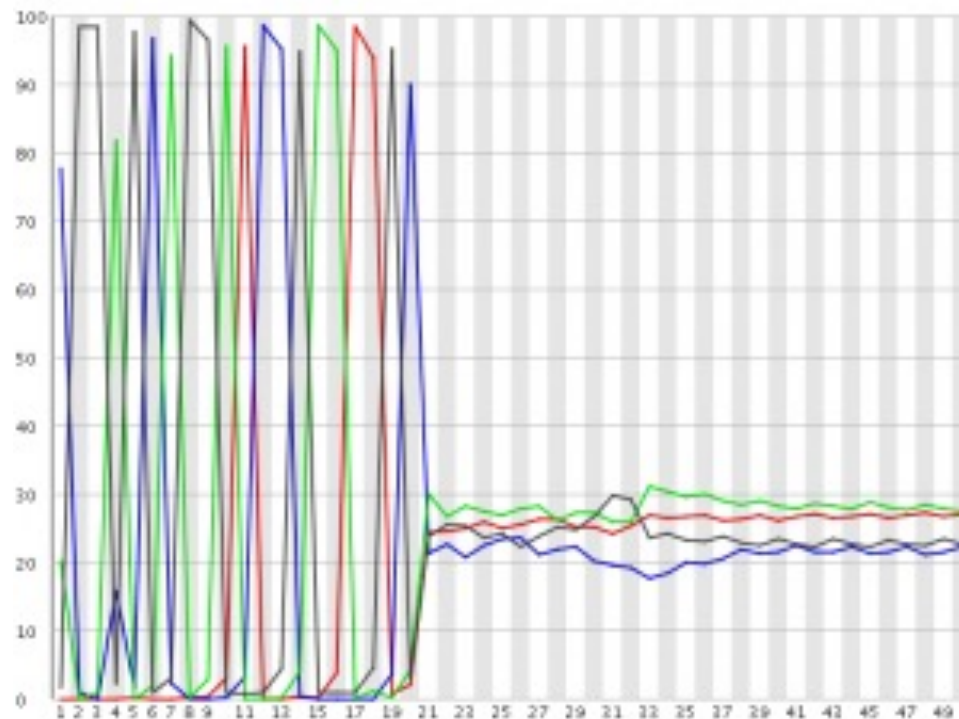
Bad: major adapter contamination,



Common Problems Identified in FASTQC

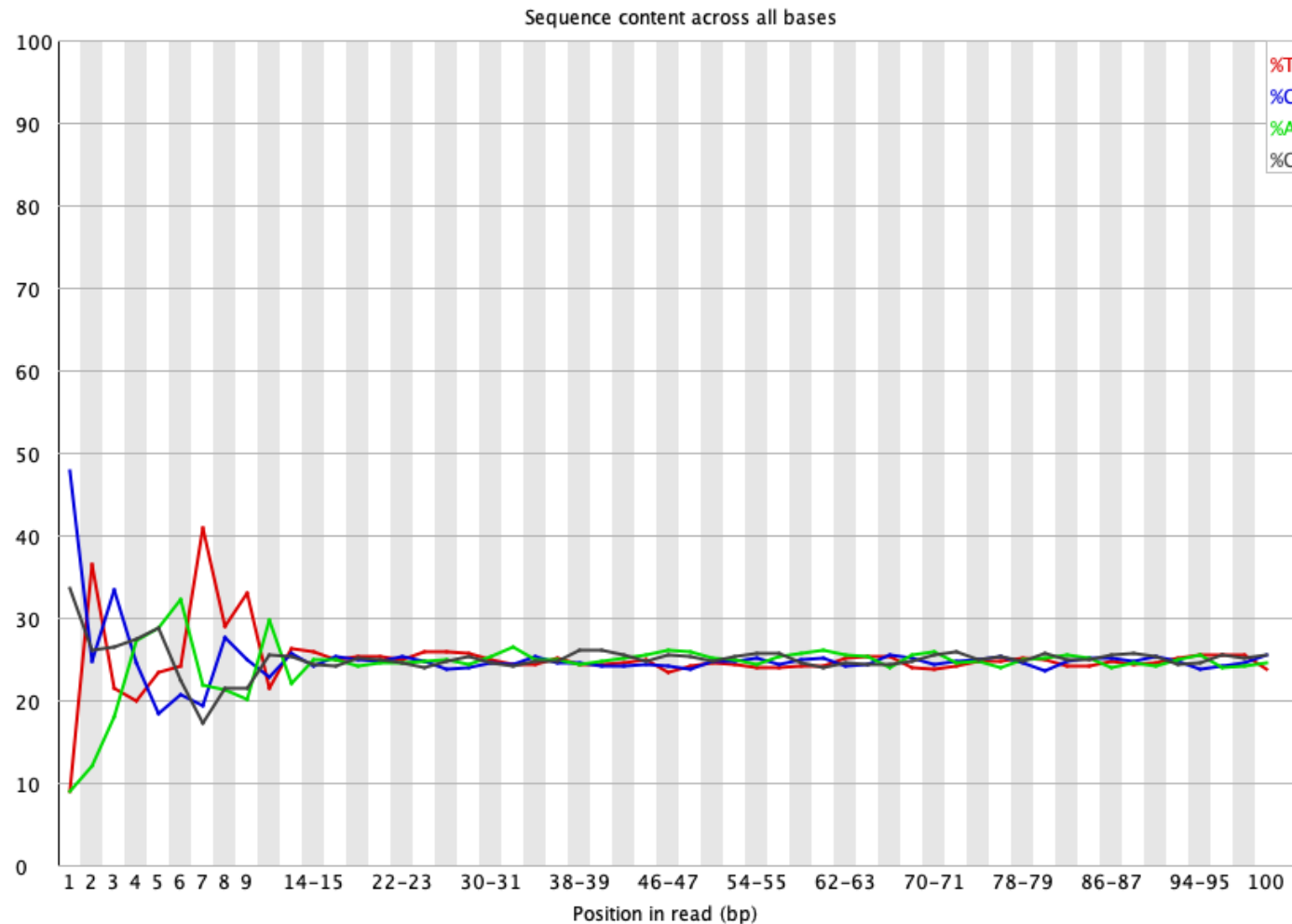
Bad: non-random beginning of sequence (e.g., sequencing common priming site), can get lots of N's or complete fail of run

Solutions: spike in diverse sequence (PhiX control is common), or use variable length primers to amplify the library (so each fragment starts at a different nucleotide)



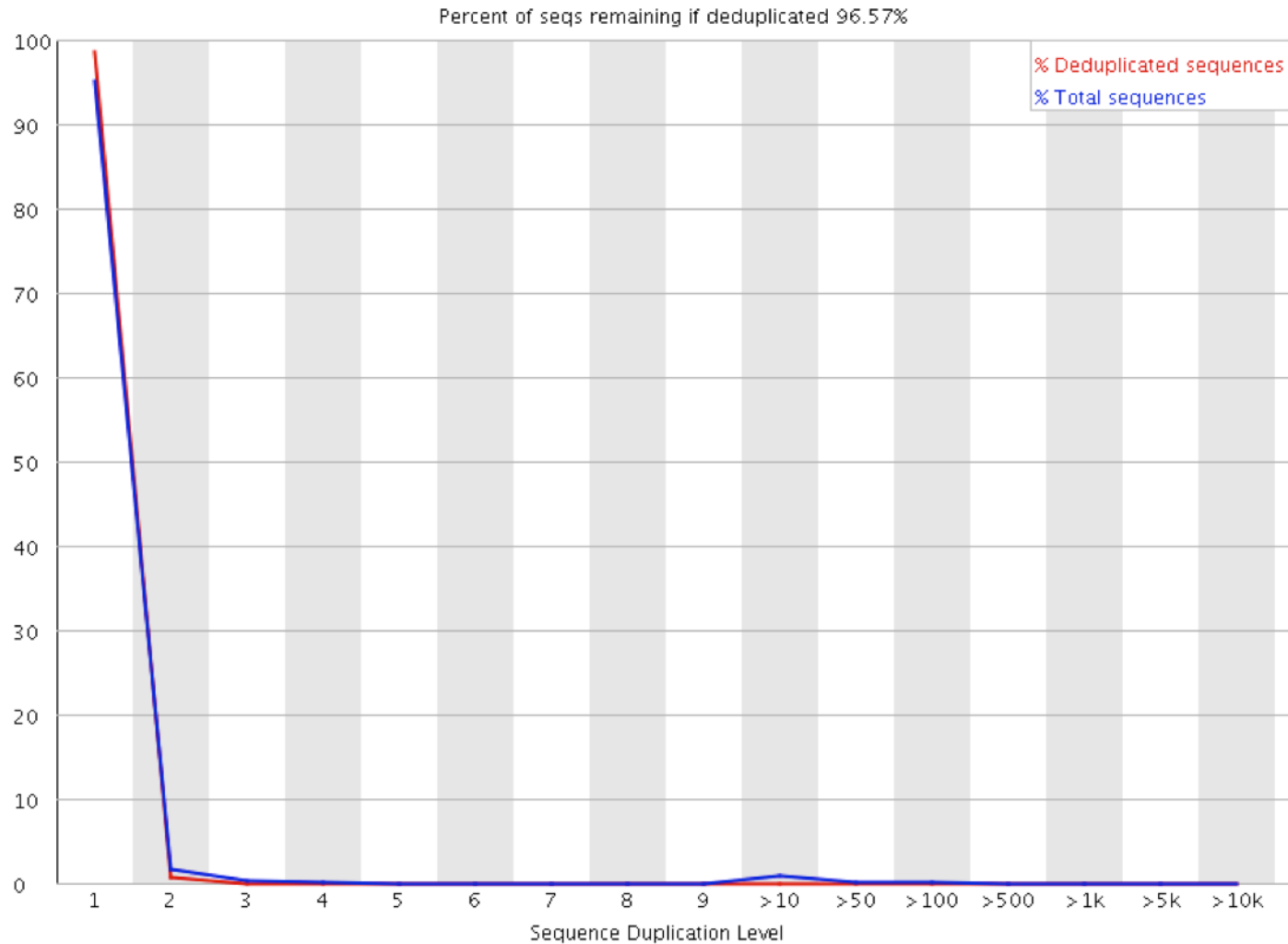
Common Problems Identified in FASTQC

Normal for RNA-seq (caused by certain “random” hexamers binding to RNA better during cDNA synthesis)



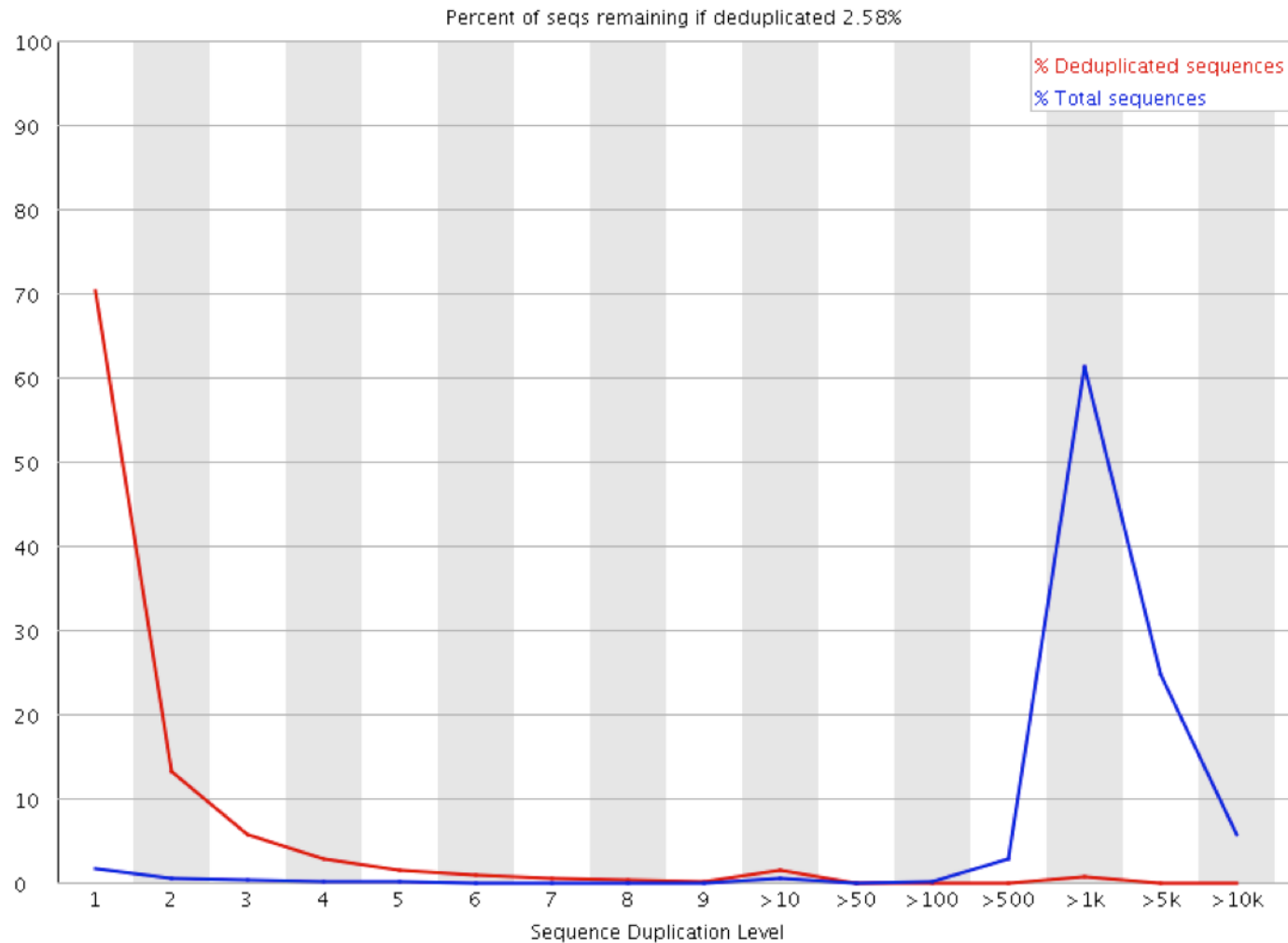
Common Problems Identified in FASTQC

Good (low sequence duplication)



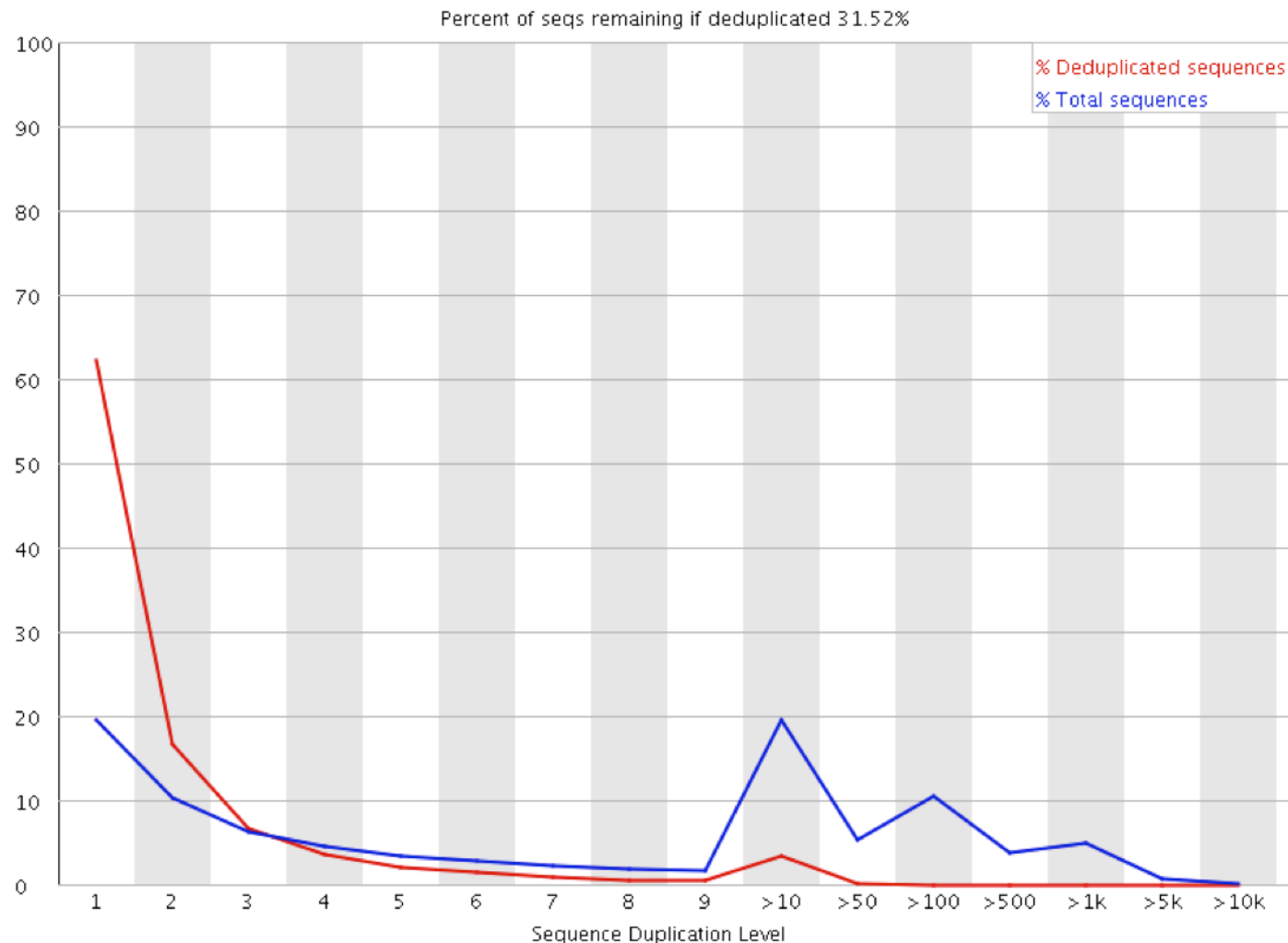
Common Problems Identified in FASTQC

Bad (very high sequence duplication)



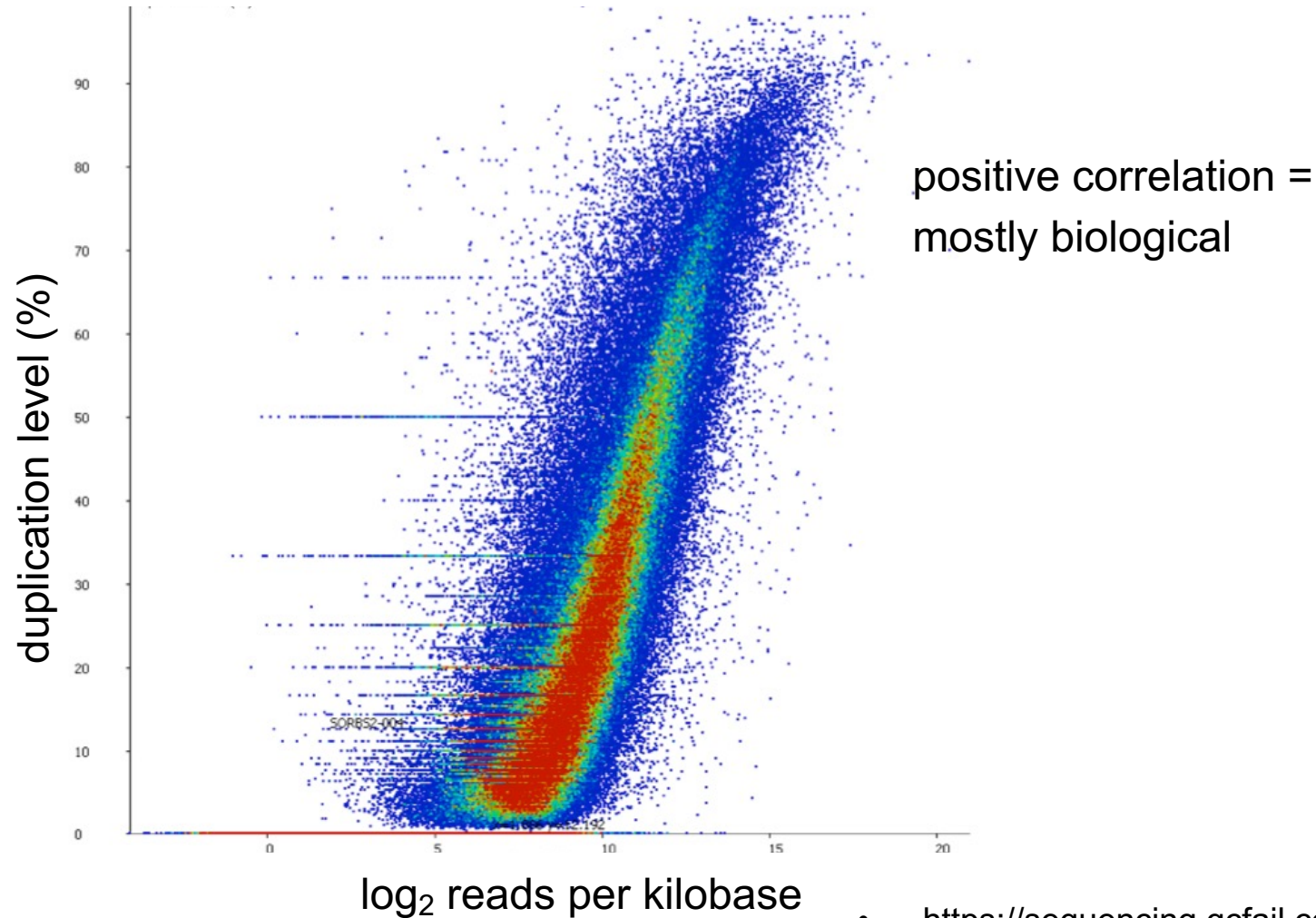
Common Problems Identified in FASTQC

Normal-ish for RNA-seq: some duplication is biological (sequencing the same parts of an RNA repeatedly)



Common Problems Identified in FASTQC

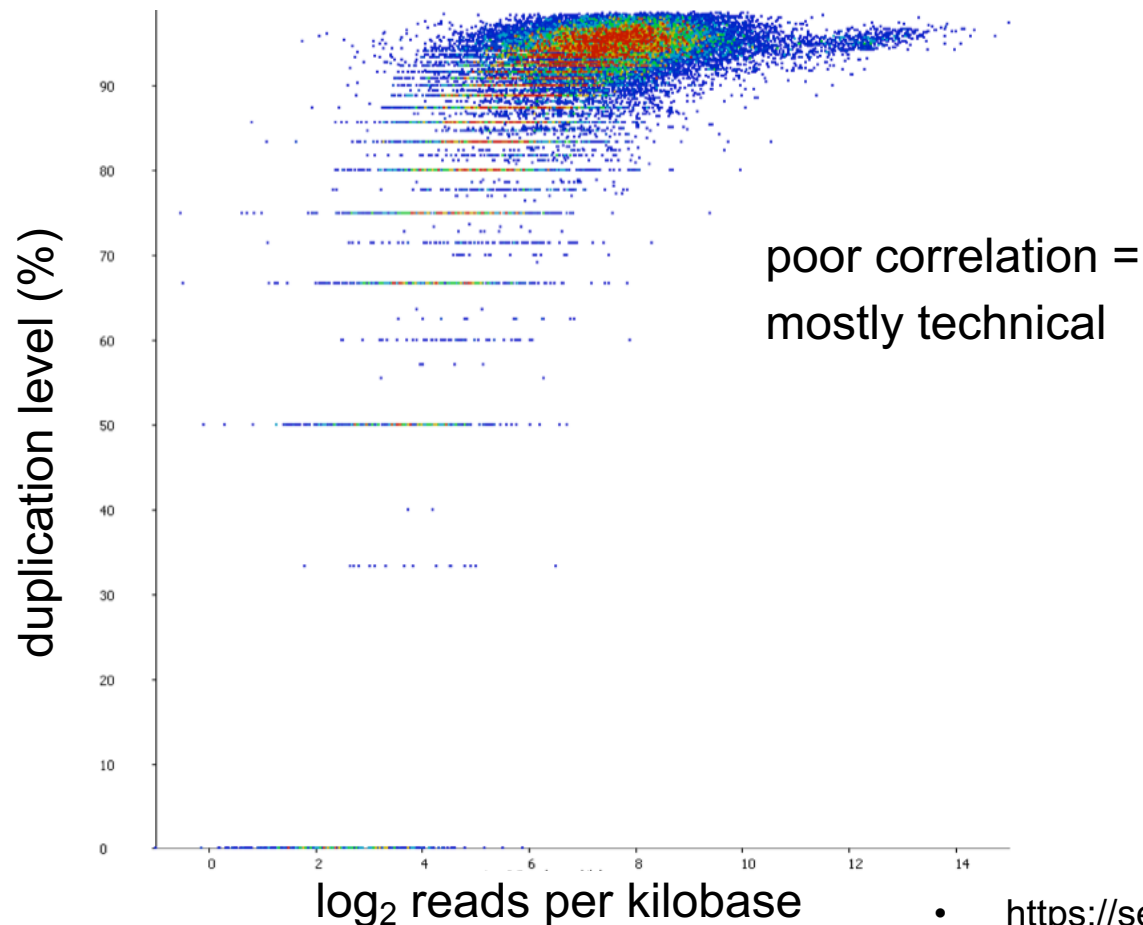
If biological, duplication level should correlate with read density in a region (e.g., highly expressed genes should have more duplicates)



- <https://sequencing.qcfail.com/articles/libraries-can-contain-technical-duplication/>

Common Problems Identified in FASTQC

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Rfastp

An Ultra-Fast and All-in-One Fastq Preprocessor (Quality Control, Adapter, low quality and polyX trimming) and UMI Sequence Parsing).

Bioconductor version: Release (3.17)

Rfastp is an R wrapper of fastp developed in c++. fastp performs quality control for fastq files. including low quality bases trimming, polyX trimming, adapter auto-detection and trimming, paired-end reads merging, UMI sequence/id handling. Rfastp can concatenate multiple files into one file (like shell command cat) and accept multiple files as input.

Author: Wei Wang [aut] , Ji-Dung Luo [ctb] , Thomas Carroll [cre, aut] 

Maintainer: Thomas Carroll <tc.infomatics at gmail.com>

Citation (from within R, enter `citation("Rfastp")`):

Wang W, Carroll T (2023). *Rfastp: An Ultra-Fast and All-in-One Fastq Preprocessor (Quality Control, Adapter, low quality and polyX trimming) and UMI Sequence Parsing*.. R package version 1.10.0.

- outputs QC in the form of base quality, GC content, duplication levels
- can also do read trimming

Trimming

Williams et al. *BMC Bioinformatics* (2016) 17:103
DOI 10.1186/s12859-016-0956-2

BMC Bioinformatics

RESEARCH ARTICLE

Open Access

Trimming of sequence reads alters RNA-Seq gene expression estimates



Claire R. Williams¹, Alyssa Baccarella², Jay Z. Parrish^{1*} and Charles C. Kim^{2,3*}

Conclusions: We find that aggressive quality-based trimming has a large impact on the apparent makeup of RNA-Seq-based gene expression estimates, and that short reads can have a particularly strong impact. We conclude that implementation of trimming in RNA-Seq analysis workflows warrants caution, and if used, should be used in conjunction with a minimum read length filter to minimize the introduction of unpredictable changes in expression estimates.

frontiers in
GENETICS

ORIGINAL RESEARCH ARTICLE

published: 31 January 2014
doi: 10.3389/fgene.2014.00013



On the optimal trimming of high-throughput mRNA sequence data

Matthew D. MacManes^{1,2*}

¹ Department of Molecular, Cellular and Biomedical Sciences, University of New Hampshire, Durham, NH, USA

² Hubbard Center for Genome Studies, Durham, NH, USA

Although very aggressive quality trimming is common, this study suggests that a more gentle trimming, specifically of those nucleotides whose PHRED score <2 or <5, is optimal for most studies across a wide variety of metrics.

Trimming

- For de novo genome or transcriptome assembly, should definitely trim adapter sequences
- For variant analyses (looking for novel mutations or SNPs), adapter and quality trimming is generally recommended
- For standard RNA-seq, newer aligners will “soft clip” from reads where ends do not match the reference
- For standard RNA-seq quality trimming (except for perhaps any “N” base calls) is generally not recommended
- If quality trimming is called for, “gentle” trimming is generally recommended (Q15 cutoff and retain reads of at least 35 bp)

My general advice: I don't think adapter trimming can possibly hurt. Be careful with quality trimming for differential expression (only do very gentle trimming if at all, maintain minimum read length)

MultiQC



- Looking at individual QC files if you have more than 10 or so samples is unwieldy
- MultiQC takes those QC file outputs and puts them in a single file
- Can look at all samples together to identify any problematic outliers